Program 1

Problem Statement
Maintain an array of Bag objects with a static count of created bags. Provide a static member function to print the total, and two friend functions that inspect the private data to report the most-sold size and the most expensive bag.

Source Code:

```
#include <iostream>
#include <string>
using namespace std;

class Bag {
private:
string name;
int price;
string sizes;

static int totalBag;

public:
// Constructor with default arguments
Bag(string n = "", int p = 0, string s = "") {
name = n;
price = p;
sizes = s;
totalBag++;
}

// Friend functions declarations
friend void findMostSold(Bag b[], int n);
friend void findExpensive(Bag b[], int n);

// Static function to show total bags
static void showTotalBag() {
cout << "Total Bags = " << totalBag << endl;
}

// Destructor
~Bag() {
}
};

// Initialize static variable
int Bag::totalBag = 0;

// Function to find most sold size
// Note: The code uses hardcoded values for sales counts (m, l, xl, xxl)
// rather than calculating from the array, based on the image.
void findMostSold(Bag b[], int n) {
int m = 35;
int l = 20;
```

```
int xl = 30;
int xxl = 25;

int max_sold = m;
string most_sold_size = "M";

if (l > max_sold) {
max_sold = l;
most_sold_size = "L";
}

if (xl > max_sold) {
max_sold = xl;
most_sold_size = "XL";
}

if (xxl > max_sold) {
max_sold = xxl;
most_sold_size = "XXL";
}

cout << "Most Sold Bag Size: " << most_sold_size << endl;
cout << "Number Sold: " << max_sold << endl;
}

// Function to find the most expensive bag
void findExpensive(Bag b[], int n) {
int max_price = b[0].price;
int index = 0;

for (int i = 1; i < n; i++) {
if (b[i].price > max_price) {
max_price = b[i].price;
index = i;
}
}

cout << "\nMost Expensive Bag:" << endl;
cout << "Name : " << b[index].name << endl;
cout << "Size : " << b[index].sizes << endl;
cout << "Price : " << b[index].price << endl;
}

int main() {
// Array of 10 bags.
// Note: This calls the constructor 10 times immediately.
Bag b[10];

for (int i = 0; i < 10; i++) {
string name, sizes;
int price;
```

```
cout << "\nEnter details of Bag " << i + 1 << endl;

cout << "Enter name: ";
cin >> name;

cout << "Enter size: ";
cin >> sizes;

cout << "Enter price: ";
cin >> price;

// Creates a temporary object and assigns it to b[i]
b[i] = Bag(name, price, sizes);
}

Bag::showTotalBag();

findMostSold(b, 10);

findExpensive(b, 10);

return 0;
}
```

Input:

Backpack
L
800
Tote
XL
600
Duffel
L
700
Satchel
M
400
Clutch
M
250
Tote
XL
850
Tote
XL
900
Duffel
L
650

Satchel
L
550
Clutch
M
250

Output:

